INSTITUTE OF SPACE TECHNOLOGY
KICSIT, Kahuta Campus

Assignment #02

Advanced Visual Programming Lab

AI Flower Classifier

Submitted By

Laiba Naeem (232201058)

BSCS-6-B

Submitted To

Sir Uzair

Submission Date

March 28, 2026

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1 Introduction

This project is an AI-based image classification system called **AI Flower Classifier**, developed using .NET and ML.NET. It classifies flower images into multiple categories using a trained machine learning model.

2 Dataset Description

Classes:

- Daisy
- Rose
- Sunflower
- Tulip
- Dandelion

Dataset Structure:

```
dataset/
  flowers/
    daisy/
    rose/
    sunflower/
    tulip/
    dandelion/
```

3 Machine Learning Model

$$X = \text{Image Features}, \quad Y = \text{Class Label}$$

$$f(X) \to Y$$

3.1 Softmax Function

$$P(y = i|x) = \frac{e^{z_i}}{\sum_{j=1}^{k} e^{z_j}}$$

3.2 ML.NET Pipeline

- Load Images
- Resize Images (224×224)
- Convert Labels to Keys
- Extract Pixels
- Train Model
- Save Model

4 Objectives

- Train ML model using dataset
- Build WinForms application
- Perform real-time prediction
- Improve accuracy

5 System Architecture

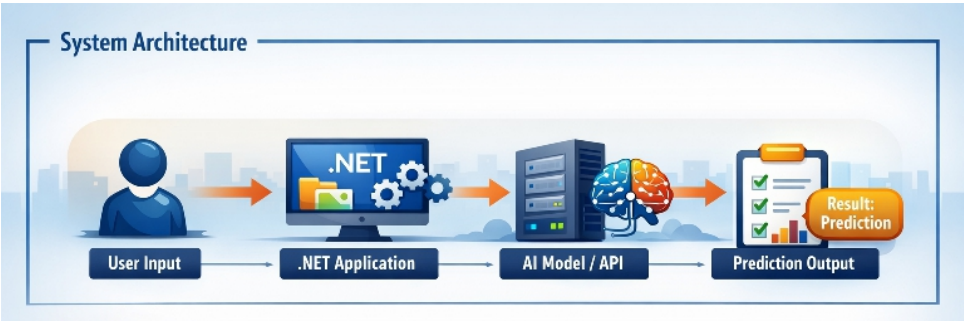

Figure 1: System Architecture

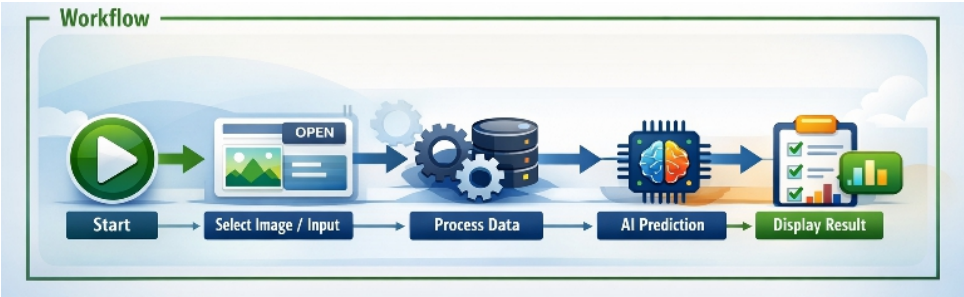

Figure 2: Workflow Diagram

6 Tools & Technologies

- .NET 7
- ML.NET
- C#
- Visual Studio
- Windows Forms

7 Code Implementation

7.1 ImageData.cs

```
namespace AI_Flower_Classifier.Data
{
    public class ImageData
    {
        public string ImagePath { get; set; } = "";
        public string Label { get; set; } = "";
    }
}
```

7.2 Model Training Code

```
var mlContext = new MLContext();

string dataPath = @"dataset/flowers";

var images = Directory.GetDirectories(dataPath)
    .SelectMany(folder => Directory.GetFiles(folder)
    .Select(file => new ImageData
    {
        ImagePath = file,
        Label = Path.GetFileName(folder)
    }));

var data = mlContext.Data.LoadFromEnumerable(images);

var pipeline = mlContext.Transforms.Conversion.MapValueToKey("
   Label")
    .Append(mlContext.Transforms.LoadImages("Image", dataPath, "
       ImagePath"))
    .Append(mlContext.Transforms.ResizeImages("Image", 224, 224))
    .Append(mlContext.Transforms.ExtractPixels("Pixels", "Image")
       )
    .Append(mlContext.MulticlassClassification.Trainers
        .LbfgsMaximumEntropy("Label", "Pixels"))
    .Append(mlContext.Transforms.Conversion.MapKeyToValue("
       PredictedLabel"));

var model = pipeline.Fit(data);
mlContext.Model.Save(model, data.Schema, "model.zip");
```

7.3 Prediction Code

```
var mlContext = new MLContext();

var model = mlContext.Model.Load("model.zip", out _);
```

```
var engine = mlContext.Model.CreatePredictionEngine<ImageData,
   ImagePrediction>(model);

var result = engine.Predict(new ImageData
{
    ImagePath = selectedImage
});

Console.WriteLine(result.PredictedLabel);
```

8 Screenshots

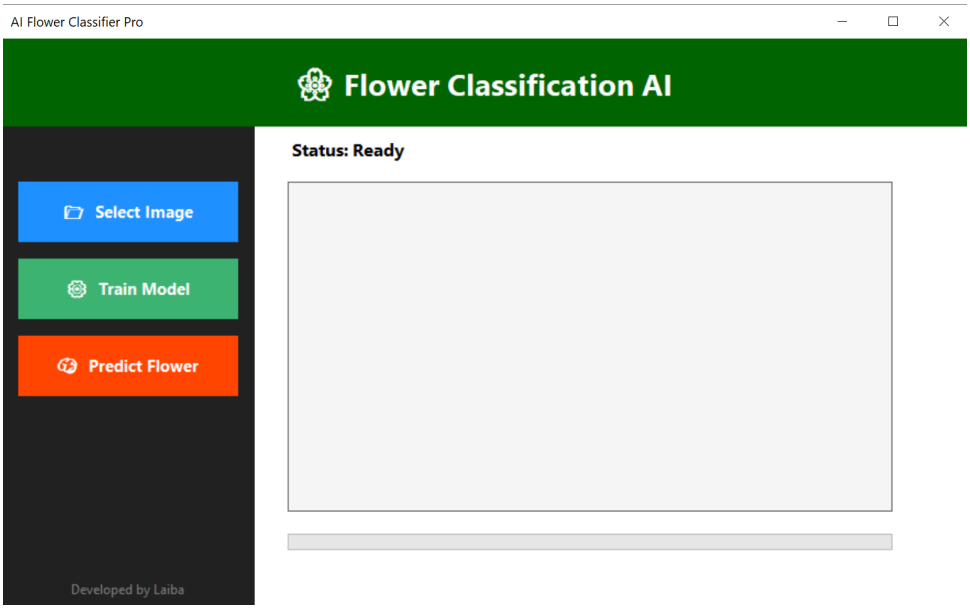

Figure 3: Main Interface

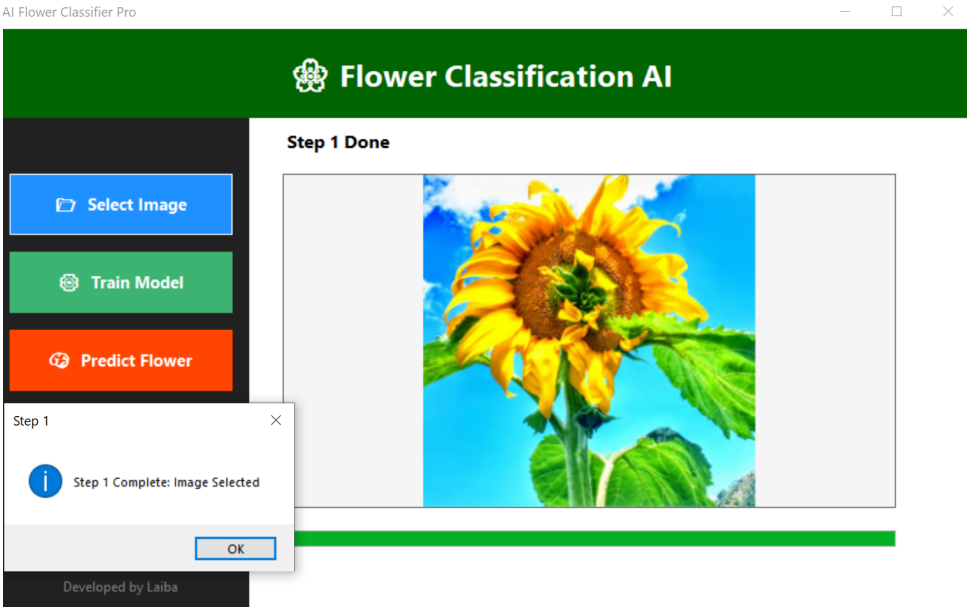

Figure 4: Image Selection

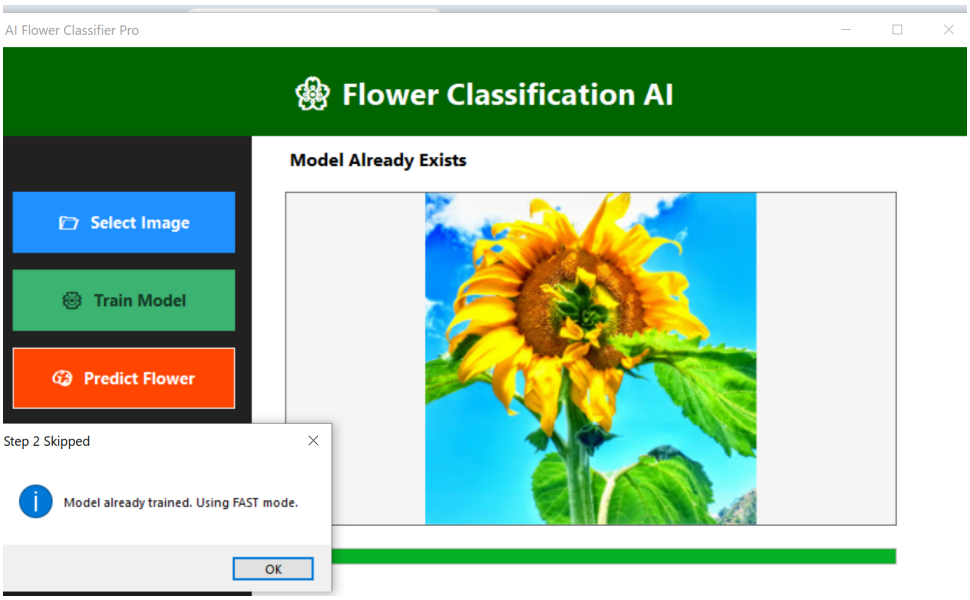

Figure 5: Training Output

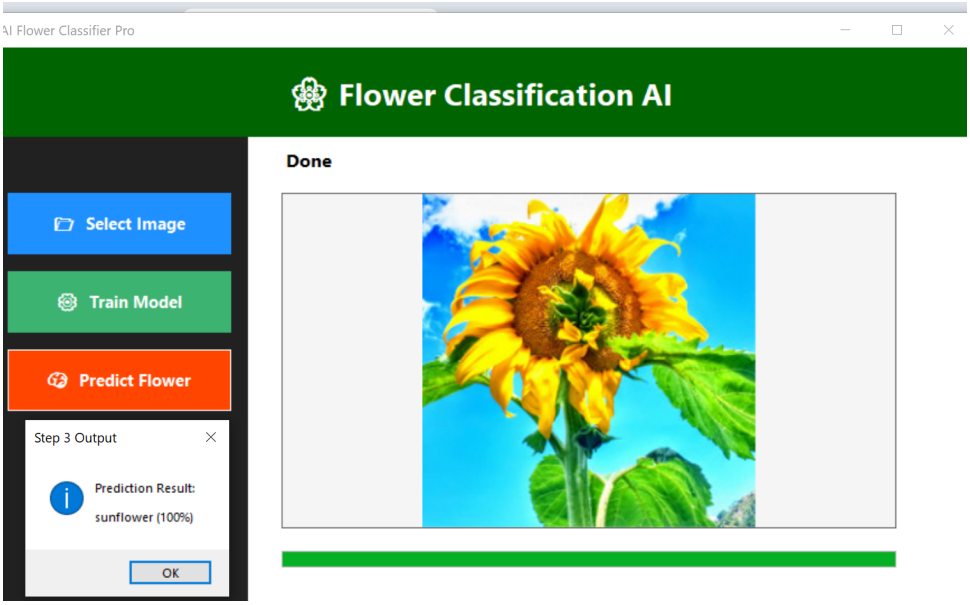

Figure 6: Prediction Result

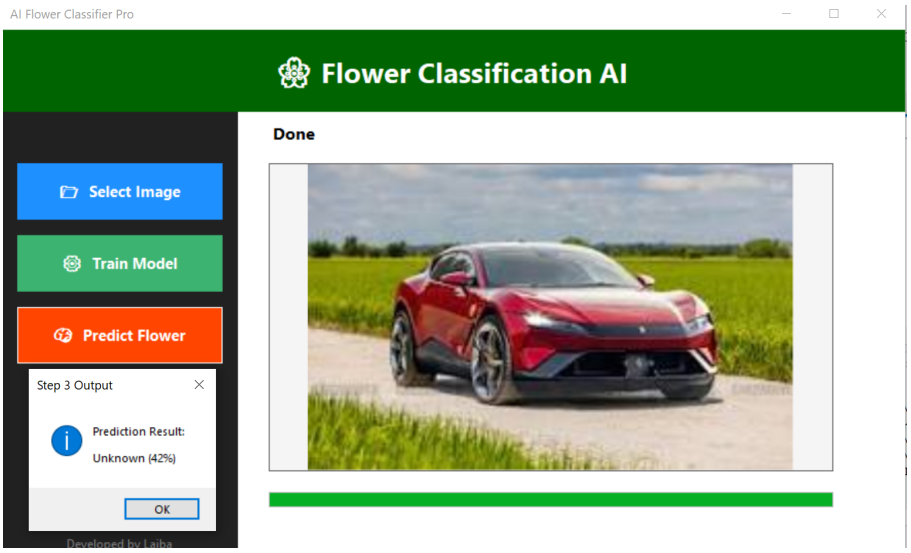

Figure 7: Validation Screen

9 Testing Report

Test Case	Input	Result
1	Daisy	Pass
2	Rose	Pass
3	Tulip	Pass
4	Sunflower	Pass
5	Dandelion	Pass

10 Results

The system successfully classifies flower images using ML.NET with good accuracy.

11 Limitations

- Small dataset
- Limited generalization

12 Future Work

- Deep learning models
- Large dataset
- Cloud deployment

13 Conclusion

This project demonstrates a complete AI-based system named **AI Flower Classifier** built using .NET and ML.NET.